Proposal overview

Investment in artificial intelligence (AI) flight simulation technology to deliver pilot training. The aim is to reduce human input and hours needed to train pilots and to significantly reduce costs of obtaining/maintaining pilot certifications.

Pilot training data

	Current (20X3)	Proposed AI training
Initial investment cost	N/A	$500 million
Time taken to train new pilots	12 months	8 months
Cost per pilot (initial training for all new pilots)	$500,000	$280,000
Cost per pilot (annual update for all pilots)	$45,000	$20,000

Training costs each year for Corjetz's pilots is currently over $140 million.

Key features of AI simulation-based pilot training

– Use of AI flight simulators to develop pilot in-flight skills, providing more realistic simulations. This reduces the need for costly and time consuming live in-flight training.

– The AI automatically collects training performance data about every aspect of individual pilot performance and behaviour. The data gathered is used to continually update the AI simulation models to improve the training scenarios.

– Instructors can use the intelligence gathered to quickly monitor the effectiveness of the training and to tailor it to individual needs.

Potential issues

– A human trainer is needed to give instructions and program the software. This is a highly specialist job and there is limited availability of instructors worldwide.

– Capability of replicating decision-making in all scenarios is still uncertain, as there are so many variables to be considered when flying a commercial aircraft.

– Regulation is moving more slowly than the technology. Due to the high safety level needed in commercial aviation, extensive testing and certifications of pilot competency will be necessary.

– Stakeholders will need to be convinced of the quality, safety and benefits of using AI.

Current successful application

One Hundrian country has invested the equivalent of over $2·5 billion in AI pilot training systems for its military pilots. Feedback has been highly positive in terms of the quality of training and cost reductions.

Summary

This is an opportunity to reduce a significant cost to the business. It is also an opportunity to gain significant competitive advantage, being the first commercial airline to introduce such cutting-edge technology.